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SUPERSEDING
Int. Fed. Spec. O–C–00746a (GSA–FSS)
June 15, 1956 and
Fed. Spec. O–C–746
May 17, 1951

FEDERAL SPECIFICATION

INHIBITOR, CORROSION, LIQUID COOLING SYSTEM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and one grade of corrosion inhibitor intended for inhibiting water and/or uninhibited ethylene glycol.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

SS–B–611—Borax (Sodium Borate).
UU–T–116 — Tape, Paper, Gummed, Water-Resistant.
PPP–B–585—Boxes: Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes, Fiber.
PPP–C–96—Cans, Metal, 28 Gage and Lighter.
PPP–T–76 — Tape, Pressure-Sensitive, Adhesive, Paper, Water Resistant.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. — The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

ASTM Standards—Part 5

ASTM Designation: D664 — Method of Test for Neutralization Value (Acid and Base Numbers) by Potentiometric Titration.

ASTM Designation: D1121—Method of Test for Reserve Alkalinity of Concentrated Engine Antifreezes.

ASTM Designation: D1287—Method of Test for pH of Concentrated Engine Antifreezes.

(Copies of ASTM Standards may be obtained from the American Society for Testing Material, 1916 Race Street, Philadelphia 3, Pennsylvania.)

Consolidated Freight Classification Ratings, Rules and Regulations.

(Copies of this publication may be obtained from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.—The corrosion inhibitor shall be suitable for the purposes intended and as specified hereinafter.

3.2 Composition.—The corrosion inhibitor shall consist of borax and mercaptobenzothiazole mixed in the proportions specified in table I.

3.2.1 *Borax.*—The borax used shall conform to the requirements of Federal Specification SS–B–611, type II.

3.2.2 *Mercaptobenzothiazole.*—The mercaptobenzothiazole used shall be of a technical grade and shall have a purity of not less than 92 percent.

TABLE I.—*Chemical composition*

Composition	Percent by weight		Test paragraph
	Minimum	Maximum	
Mercaptobenzothiazole	3.5	4.5	4.4.2.1
Borax	95.5	96.5	4.4.2.2

3.3 Other chemical requirements.—In addition to conforming to the requirements of 3.2, 3.2.1, and 3.2.2, the corrosion inhibitor shall conform to the additional chemical requirements specified in 3.3.1 to 3.3.4, inclusive.

3.3.1 *pH value.*—The initial pH of the corrosion inhibitor when dissolved in ethylene glycol shall be 5.8 to 6.3. When diluted with distilled water, the pH shall be 7.5 to 8.0 when tested as specified in 4.4.2.3.

3.3.2 *Reserve alkalinity.*—The reserve alkalinity shall be not less than 12.0 when tested as specified in 4.4.2.4.

3.3.3 *Insoluble matter.*—The insoluble matter shall be not greater than 0.5 percent by weight when tested as specified in 4.4.2.5.

3.3.4 *Chlorides, sulfates, and carbonates.*—The corrosion inhibitor shall contain no chlorides, sulfates, or carbonates when tested as specified in 4.4.1.1 and 4.4.1.2.

3.4 Stability.—Stability shall be such that the corrosion inhibitor will conform to this specification after storage in the original unopened containers for a 12-month period under any climatic conditions.

3.5 Workmanship.—The corrosion inhibitor shall be processed in accordance with high-grade commercial practice to produce a uniform product and shall be free of dirt and grit.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—A lot shall consist of all the corrosion inhibitor produced in a single batch and offered for acceptance at one time.

4.2 **Sampling.** — Two containers shall be taken by the contractor at random from among the containers comprising the lot for testing as specified in 4.3 and 4.4.

4.3 **Inspection tests.**—The contractor shall furnish all samples and shall be responsible for accomplishing the tests specified herein. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the service of a commercial testing laboratory acceptable to the inspector. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4 **Tests.**

4.4.1 *Qualitative tests.*—The corrosion inhibitor shall meet the test requirements as specified in 4.4.1.1 through 4.4.1.4.

4.4.1.1 *Chlorides.*—Dissolve 0.25 gm. of the corrosion inhibitor in 10 ml. of warm 50/50 ethylene glycol (chloride free) and distilled water mixture. Add 10 drops of concentrated nitric acid, agitate, and filter. To the filtrate add four drops of a 3-percent aqueous solution of silver nitrate. No precipitate shall form.

4.4.1.2 *Sulfate and carbonates.*—Dissolve 0.25 gm. of the corrosion inhibitor in 10 ml. of warm 50/50 ethylene glycol and distilled water mixture. Add 1 ml. of 5-percent aqueous solution of barium chloride. No precipitate shall form.

4.4.1.3 *Borate.*—To 0.25 gm. of the corrosion inhibitor add 10 ml. of ethylene glycol and place in a 500-milliliter Erlenmeyer flask. Warm and stir until dissolved. Add 10 ml. of anhydrous methanol and 10 ml. of concentrated sulphuric acid. Fit a one-hole rubber stopper into the flask. Insert a 6-inch piece of 6-mm. O.D. glass tubing so that one end extends 6 mm. below the bottom of the rubber stopper. Extend the other end two inches into a 4-inch piece of 12-mm. O.D. glass tubing serving as a test chimney. The assembled apparatus is placed upon a wire gauze which covers a heat source. The flask is then heated to boiling. The vapors escaping from the chimney shall be ignited. The ignited vapors shall burn with a green-colored flame.

4.4.1.4 *Mercaptobenzothiazole.* — To 0.25 gm. of the corrosion inhibitor add 10 ml. of ethylene glycol and 10 ml. of distilled water. Warm and stir until dissolved. Add 5 ml. of 5 percent benzene solution of copper oleate. Shake and allow to stand for 5 minutes. A butter-yellow precipitate shall develop in the benzene layer and upon standing will be located between the benzene layer and the water layer.

4.4.2 *Quantitative tests.*

4.4.2.1 *Mercaptobenzothiazole.* — Dry approximately 15 gm. of corrosion inhibitor at 70° C. for 2 hours. Extract 5 gm. of the dried inhibitor which has been weighed to plus or minus 0.1 mg. in a Soxhlet apparatus with acetone until the drainings are colorless. Evaporate the acetone extract on a steam bath, dissolve the residue in 20 ml. hot benzene, filter, wash any insoluble matter five times with 10-milliliter portions of warm benzene. To the combined filtrate and washings contained in a 100-milliliter centrifuge tube add an excess of a 5-percent filtered solution of copper oleate in benzene so that

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the solution remains green. Let stand overnight, centrifuge the orange-yellow precipitate, wash two times with benzene and then one time with petroleum ether while in the centrifuge tube. Dry to constant weight at 140° C., place in desiccator; when cool, weigh rapidly.

Note.—The copper salt of mercaptobenzothiazole which forms at first and which is soluble in benzene changes to the insoluble cuprous salt and benzothiazyl disulfide (a change similar to the decomposition of cupric cyanide to cuprous cyanide and cyanogen). The precipitate contains, therefore, exactly one-half the mercaptobenzothiazole present. Its weight represents 68.7 percent; therefore it is to be multiplied by a factor of 1.4556.

4.4.2.2 *Borax.*—Dry approximately 15 gm. of the corrosion inhibitor for 48 hours in a desiccator over anhydrous $CaCl_2$. Weigh approximately 10 gm. to the nearest tenth of a milligram and dissolve in 300 ml. of freshly boiled distilled water. Filter into a 500-milliliter volumetric flask. Wash residue with three 50-milliliter portions of boiled distilled water. Cool contents to 20° C. and dilute to the mark with cooled boiled distilled water. Take a 100-milliliter aliquot for analysis. Each aliquot contains one-fifth of the sample weight.

Titrate the 100-milliliter aliquot with standardized 1*N* HCl to a pH of 4.2 using glass electrodes and instrument described in method ASTM Designation: D 664–54. Calculate percent of alkali as borax. One milliequivalent of 1*N* HCl = 0.1908 grams $Na_2B_4O_7.10H_2O$.

$$\text{Percent alkali (borax)} = \frac{(\text{ml. of HCl})\ (\text{normality of HCl})\ (0.1908)\ (5)\ (100)}{\text{weight of sample}}$$

Bring the titrated sample to a boil and continue boiling for 1 minute. Add 50 ml. of glycerol which has been neutralized to a pH of 7.0 with 0.1*N* NaOH. Titrate with standardized 1*N* NaOH to a pH of 9.0 using glass electrodes and instrument described in method ASTM Designation: D 664–54. Add 10 ml. of neutralized glycerol. If no change in pH is observed, titration is complete. If a change occurs, add 10 milliliter portions of the glycerol and subsequently titrate to a pH 9.0 until no change occurs. Calculate the percent acid. One milliequivalent of NaOH = 0.0954 grams of $Na_2B_4O_7.10H_2O$.

$$\text{Percent acid} = \frac{(\text{ml. NaOH})\ (\text{normality of NaOH})\ (0.0954)\ (5)\ (100)}{\text{weight of sample}}$$

The difference between the alkali and acid percentage shall be not more than 2.0 percent and neither value shall be less than 95.5 percent.

4.4.2.3 *pH value.*—The pH determination shall be made at 70° to 80° F. using the apparatus and procedure specified in ASTM Designation: D 1287.

4.4.2.3.1 *Preparation of solution.*—To 1.0 gram of the corrosion inhibitor add 40 ml. of ethylene glycol which has been neutralized to a pH of 7.0 ± 0.1, and 2.0 ml. of distilled water. Warm and stir until the compound is dissolved and cool. The initial pH shall be measured at this point. To the solution prepared above add 80 ml. of distilled water and again determine the pH.

4.4.2.4 *Reserve alkalinity.* — The reserve alkalinity shall be determined in accordance with ASTM Designation: D 1121.

4.4.2.4.1 *Preparation of solution.*—Weigh 0.25 gm. ± 0.001 gm. of the corrosion inhibitor. Add 100 ml. of distilled water. Warm and stir until dissolved.

4.4.2.5 *Insoluble matter.*—Dissolve 10 gm. of the corrosion inhibitor in 900 ml. of a mixture of equal volumes of ethylene glycol and distilled water. Warm and stir. Filter through a suitable tared filtering crucible, rinse the container thoroughly with 5 ml. of the 50/50 glycol-water mixture, completely transferring all insoluble matter to the crucible. Wash with five 15-milliliter portions of

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the glycol-water mixture and dry to constant weight at 105° C. Gain in weight is insoluble matter.

4.5 Rejection and retest.—If either test sample fails any of the tests the lot shall be rejected. However, the contractor may remove or rework nonconforming material and resubmit the lot for acceptance.

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.*—The corrosion inhibitor shall be packaged in 5-ounce metal cans conforming to the requirements of Federal Specification PPP–C–96, type V, class 1. Unless otherwise specified, exterior coating of the cans shall comply with paragraph 3.3.2.2 of Federal Specification PPP–C–96.

5.1.1.2 *Intermediate packaging.* — Twenty four unit cans of the corrosion inhibitor, packaged as specified in 5.1.1.1, shall be packaged in close fitting intermediate containers conforming to Federal Specification PPP–B–636. Closure shall be in accordance with the appendix to the box specification. Tape for closure shall conform to Federal Specification UU–T–116 or PPP–T–76.

5.1.2 *Level B.*

5.1.2.1 *Unit packaging.*

5.1.2.1.1 *Military agencies.*—The unit packaging shall conform to the requirements specified in 5.1.1.1.

5.1.2.1.2 *Civilian agencies.*—The corrosion inhibitor shall be packaged in an 8½-ounce, string opening, cylindrical pressed board container equipped with metal ends, crimped. After crimping, the edge of the crimp shall be smooth, without excess gathering of metal or paper. The cover shall be furnished with metal ends crimped and provided with a skirt that can be slipped over the open end of the container and shall be tight enough not to fall off by its own weight when empty container is held in an inverted position.

5.1.2.2 *Intermediate packaging.* — Twenty four unit containers of the corrosion inhibitor, packaged as specified in 5.1.2.1.1 or 5.1.2.1.2, shall be packaged in close fitting fiberboard boxes conforming to Federal Specification PPP–B–636. Closure shall be made in accordance with the appendix to the box specification.

5.1.3 *Level C.* — Unit and intermediate packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Four intermediate packages, packaged as specified in 5.1.1.2, shall be packed into snug fitting exterior boxes conforming to Federal Specification PPP–B–621, class 2; PPP–B–601, overseas type; or PPP–B–636, type I or II, grade 5. Closures and strapping shall be in accordance with the applicable box specification.

5.2.2 *Level B.*—Four intermediate packages, packaged as specified in 5.1.2.1.1 or 5.1.2.1.2, shall be packed into snug fitting exterior boxes conforming to Federal Specification PPP–B–636, type I or II, class 1; PPP–B–585, class 1; PPP–B–591, domestic type; or PPP–B–621, class 1. Closure and strapping shall be in accordance with the applicable box specification.

5.2.3 *Level C.*—Corrosion inhibitor, packaged as specified in 5.1.3, shall be packed into commercial containers which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Contain-

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ers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civilian agencies.*—Marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required in the contract or order, marking of the interior packages and shipping containers shall be in accordance with Military Standard MIL–STD–129. The nomenclature shall be "Inhibitor, Corrosion, Liquid Cooling System".

6. NOTES

6.1 Intended use.—Corrosion inhibitor compound, as specified herein, is intended for use in the cooling system of liquid-cooled internal combustion engines; specifically to inhibit water and/or uninhibited ethylene gylcol-type antifreeze. This corrosion inhibitor shall not be used in the cooling system of liquid cooled aircraft engines.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Transportation description.—Transportation description applicable to this item is:

Chemicals, not otherwise indexed by name (released to valuation of $.50 per pound).
Carload minimum weight 30,000 pounds.
Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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